Thermopolis Graft Protocol

Urban climate intelligence.

Analyze any street. Understand heat. Design cooler cities.

Calle de Fuencarral →

Madrid	Lyon	Naples	Lisbon	Abidjan	Tokyo	Phoenix	Singapore
Spain	Coming soon	Coming soon	Coming soon	Coming soon	Coming soon	Coming soon	Coming soon

Calle de Fuencarral

Madrid, Spain

MOD +

URBAN RANKING

AVERAGE

Among the **55% hottest** streets in Madrid · -0.1°C vs city average

THERMAL SCORE	SURFACE TEMP	LENGTH	ORIENTATION	WIND COMFORT	SOLAR EXPOSURE
45.2 /100	39.6 °C	2427.1 m	66.7 °	71.2 /100	High
Moderate	Avg. July afternoon	Analysis corridor	Relative to sun path	Good	Score 81.6/100

STREET VERDICT

MODERATE

Composite reading combining projected heat, canyon geometry, solar exposure, ventilation, vegetation and social vulnerability.

STREET TREES

Vegetation layer

Recorded	Spacing	Main species
431	6.8 m	Acacia del Japón/Sophora japonica, Aligustre del Japón, Ciruelo púrpura

CANYON RATIO

1.38

Avg. height 16.6m · width ~12m

SOCIAL VULNERABILITY — SECTION 2807901075

Thermal justice context

Population	Aged 65+	Elderly 2050	Income
~1214	14.7%	~239	~57 171€

MODERATE 44.1/100 Moderate thermal stress with medium social vulnerability · avg. across 6 sections.

Source: INE Censo Anual 2022 · INE Atlas Renta 2021 · estimates based on census section data

PHASE 02 — MOTHER GRAFT NETWORK

Graft network

A compact spatial reading of graft anchors, cooling outlets and thermal continuity.

MOTHER GRAFTS

25

available anchors

COOLING OUTLETS

30

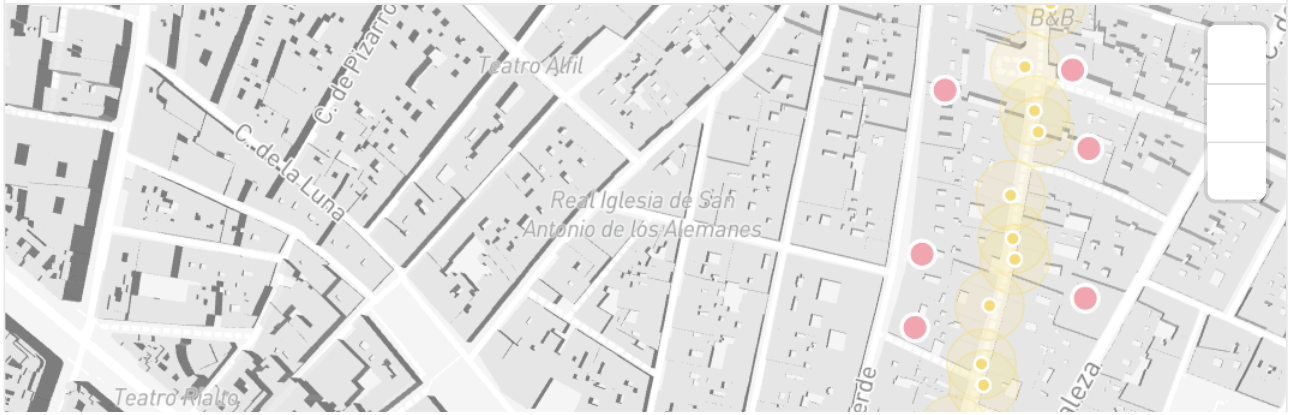
83m · 60s walk

INTERACTIVE MAP

Graft spatial distribution

2D 3D

Cooler GP Producer GP Cooling outlet · 83m



COVERAGE

100%

continuous protection



ESTIMATED FELT TEMPERATURE REDUCTION

-8 to -12°C

at outlet level · 7m pedestrian radius

COVERAGE

100%

continuous corridor

COOLING OUTLETS

30

83m · 60s walk

PEDESTRIAN CONTINUITY

Restored

thermal relay active

THERMOSIPHON CIRCUIT

Closed-loop cooling system

closed loop — thermosiphon — no pump

COOLER GP

PRODUCER GP

SIDES

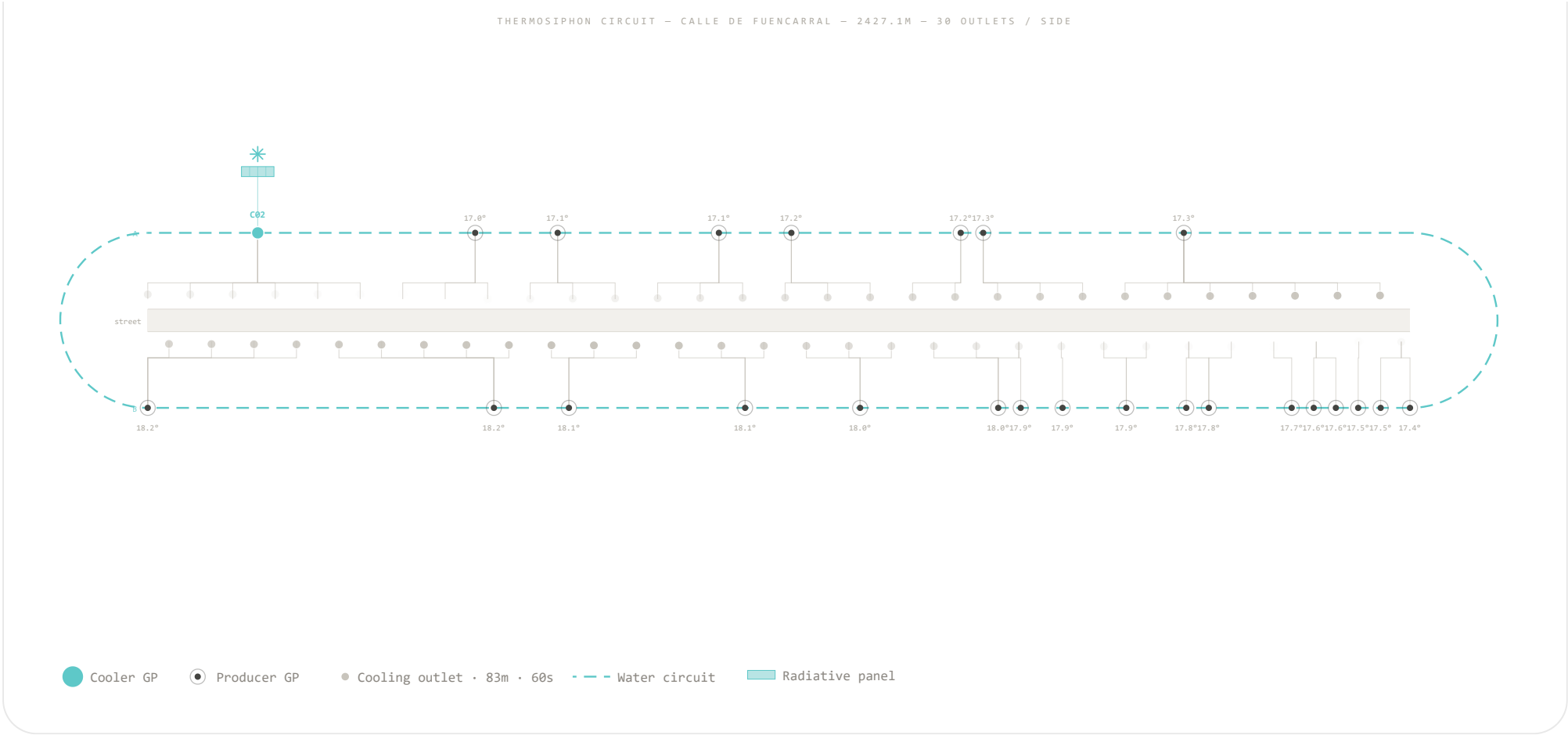
PANEL AREA

1	24	8 / 17	316.3m²
radiative panels	heat absorbers	side A / side B	total area

8 GP side A (west→east) + 17 GP side B (east→west) = closed perimeter loop

COOLING CAPACITY	RADIATIVE PANEL TOTAL	AVG. SKY VIEW FACTOR
20.8 kW	316.3 m²	0.888

CIRCUIT DIAGRAM



GRAFT POINT DETAILS

Point catalogue

GP-01 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

INTERVENTION STRATEGY

Resident comfort + street cooling — dual benefit

If occupied terrace: mother graft doubles as shading canopy — resident benefit

BUILDING A — LOWER

3m

1 floors

RESIDENTIAL

BUILDING B — HIGHER

24m

8 floors

RESIDENTIAL

COOLING OUTLET EFFECT

-3.7 to -7.3°C felt

Estimated felt temperature reduction at street level.

TECHNICAL DATA

TYPE	VERTICAL_GAP
PRIORITY	HIGH



POTENTIAL	VERY HIGH
GRAFT SYSTEM	Mother Graft
HEIGHT GAP	21m
ANCHOR HEIGHT	3m roofline
COORDINATES	40.43344N, 3.70489W

unknown — site verification required

Open in Google Maps →



THERMAL ANALYSIS

THERMAL ISSUES

Vertical discontinuity of 21.0m between adjacent buildings. The height gap allows solar radiation to penetrate at angles that maximize thermal absorption on lower facades and pedestrian level.

ANCHOR CONDITIONS

Structural gap detected: Height rupture 21.0m — anchor at 3.0m — facade 394.8m² — SVF 0.988 — side B. Both facades offer anchor points at parapet level. The vertical discontinuity creates a natural chimney effect that can be harnessed for passive cooling.

SOCIAL SIGNALS

Height transition zones are underactivated urban spaces. Pedestrian flow is typically unshaded at this junction point, creating thermal stress during peak hours.

GRAFT POTENTIAL

EXCELLENT — Symbiotic membrane spanning 21.0m gap can reduce felt temperature by 8-12°C. Dual-facade anchoring enables tensile structure with integrated misting or green wall system.

GP-02 VERTICAL_GAP MEDIUM GOOD

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

INTERVENTION STRATEGY

Resident comfort + street cooling — dual benefit

If occupied terrace: mother graft doubles as shading canopy — resident benefit

BUILDING A — LOWER

18m

6 floors

RESIDENTIAL

BUILDING B — HIGHER

24m

8 floors

UNKNOWN

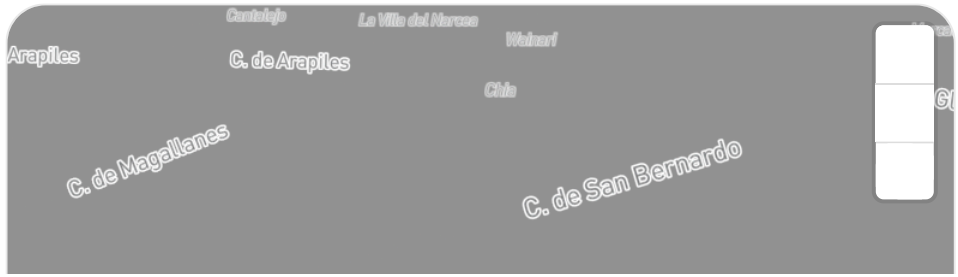
COOLING OUTLET EFFECT

-2.2 to -4.4°C felt

Estimated felt temperature reduction at street level.

TECHNICAL DATA

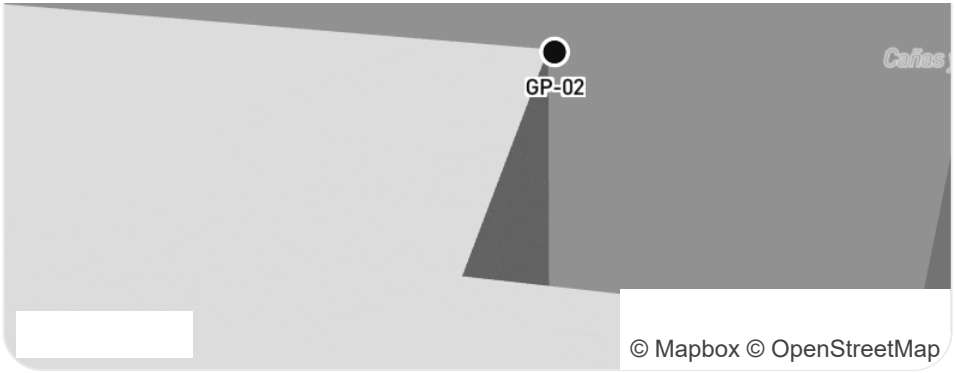
TYPE	VERTICAL_GAP
PRIORITY	MEDIUM



POTENTIAL	GOOD
GRAFT SYSTEM	Mother Graft
HEIGHT GAP	6m
ANCHOR HEIGHT	18m roofline
COORDINATES	40.43248N, 3.70447W

unknown — site verification required

Open in Google Maps →



THERMAL ANALYSIS

THERMAL ISSUES

Vertical discontinuity of 6.0m between adjacent buildings. The height gap allows solar radiation to penetrate at angles that maximize thermal absorption on lower facades and pedestrian level.

ANCHOR CONDITIONS

Structural gap detected: Height rupture 6.0m — anchor at 18.0m — facade 126.6m² — SVF 0.761 — side A. Both facades offer anchor points at parapet level. The vertical discontinuity creates a natural chimney effect that can be harnessed for passive cooling.

SOCIAL SIGNALS

Height transition zones are underactivated urban spaces. Pedestrian flow is typically unshaded at this junction point, creating thermal stress during peak hours.

GRAFT POTENTIAL

EXCELLENT — Symbiotic membrane spanning 6.0m gap can reduce felt temperature by 8-12°C. Dual-facade anchoring enables tensile structure with integrated misting or green wall system.

GP-03 VERTICAL_GAP HIGH VERY HIGH

Public space activation — civic thermal infrastructure

Mother graft — vertical anchor

+

GP-04 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-05 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-06 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-07 VERTICAL_GAP LOW LOW

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-08 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-09 VERTICAL_GAP HIGH HIGH

Urban thermal corridor — general street cooling

Mother graft — vertical anchor

+

GP-10 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-11 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-12 VERTICAL_GAP HIGH HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-13 VERTICAL_GAP MEDIUM GOOD

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-14 VERTICAL_GAP HIGH HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-15 VERTICAL_GAP LOW LOW

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-16 VERTICAL_GAP MEDIUM GOOD

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-17 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

+

GP-18 VERTICAL_GAP HIGH HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-19 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-20 VERTICAL_GAP HIGH VERY HIGH

Daytime thermal comfort — office hours priority

Mother graft — vertical anchor

GP-21 VERTICAL_GAP HIGH HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-22 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-23 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-24 VERTICAL_GAP HIGH VERY HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

GP-25 VERTICAL_GAP HIGH HIGH

Resident comfort + street cooling — dual benefit

Mother graft — vertical anchor

PHASE 03

Architectural prescription

AI-generated spatial response translating the thermal diagnosis into a graft intervention strategy.

DIAGNOSIS

Calle de Fuencarral is projected to experience a critical thermal environment by 2050, with estimated temperatures soaring to 39.6°C. The street's high solar exposure (81.6/100) combined with insufficient vegetation (10/100) and a moderate thermal score (54.8/100) indicates a significant risk of urban heat island effects. This necessitates immediate architectural interventions to enhance livability and sustainability for future residents and visitors.

PRIORITY GRAFTS

1 **Point 1 — Vertical Gap**

- **Location**: 18.0m anchor, facade 126.6m², side A

- **Why**: Medium priority due to potential for improved shading and air circulation.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Community rooftop gardens to enhance social interaction and biodiversity.

2 **Point 2 — Vertical Gap**

- **Location**: 3.0m anchor, facade 428.2m², side A

- **Why**: High priority for maximizing vertical shading and reducing heat accumulation.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Public art installations to engage the community.

3 **Point 3 — Vertical Gap**

- **Location**: 3.0m anchor, facade 299.7m², side A

- **Why**: High priority for enhancing thermal comfort through effective shading.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Urban seating areas with greenery for relaxation.

4 **Point 5 — Vertical Gap**

- **Location**: 9.0m anchor, facade 509.4m², side A

- **Why**: High priority to create a cooling microclimate along the street.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Create shaded walkways to encourage pedestrian activity.

5 **Point 6 — Vertical Gap**

- **Location**: 3.0m anchor, facade 408.8m², side A

- **Why**: High priority due to its significant facade area for shading.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Vertical gardens to improve air quality and aesthetics.

6 **Point 8 — Vertical Gap**

- **Location**: 3.0m anchor, facade 573.8m², side A

- **Why**: High priority for maximizing the cooling effect in a densely built environment.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Interactive community panels for education on environmental issues.

7 **Point 10 — Vertical Gap**

- **Location**: 15.0m anchor, facade 326.5m², side B

- **Why**: High priority to enhance visual interest and thermal efficiency.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Community workshops and events to foster neighborhood connections.

8 **Point 24 — Vertical Gap**

- **Location**: 3.0m anchor, facade 612.9m², side B

- **Why**: High priority due to its significant facade area for reducing solar gain.

- **Type of Graft**: Mother Graft — vertical anchor

- **Social Program**: Outdoor markets or gathering spaces for local vendors.

CORRIDOR

Calle de Fuencarral serves as a vital corridor linking Casa de Campo to the Centro, facilitating pedestrian flow and connectivity. The proposed grafts will not only enhance thermal comfort but also reinforce the street's role as a social and cultural artery. By integrating green infrastructure and public amenities, this corridor will promote sustainable urban living while mitigating the impacts of rising temperatures.

WARNING

If the thermal coverage remains insufficient despite the proposed grafts, alternative strategies must be considered. These could include the implementation of cool roofs, reflective pavements, and increased tree canopy coverage throughout the corridor. Additionally, the integration of water features or misting systems can provide immediate cooling relief to pedestrians during peak heat periods. Engaging local communities in green initiatives can further enhance the overall resilience of Calle de Fuencarral against climate challenges.